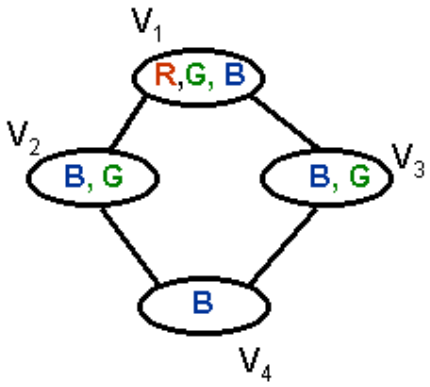
Tutorial #9: Constraint Satisfaction Problem

Simple Map Colouring Problem

Let's look at a simple map coloring problem and explore the behavior of constraint propagation, backtracking and forward checking in this context. Here is the relevant problem specification:



1. The type of graph shown above is known as a _____ graph for the CSP.
2. Write the values in each of the indicated variable domains **after** any changes required to achieve arc consistency for just that arc. Then, assume that the following arcs are done sequentially, with the effects on the domains propagating. Enter domains as a sequence of letters, for example, `R B`, do not include parenthesis, do not spell out colour names, leave spaces between the letters. If there are no values left in a domain, enter `none`.

V_1 - V_2: D_1= _____ D_2= _____
V_1 - V_3: D_1= _____ D_3= _____
V_2 - V_4: D_2= _____ D_4= _____
V_3 - V_4: D_3= _____ D_4= _____
V_1 - V_2: D_1= _____ D_2= _____
V_1 - V_3: D_1= _____ D_3= _____

3. Assume we use pure backtracking to search for a solution to this problem. We use a fixed variable ordering in the search (V_1, V_2, V_3 and V_4) and the values are considered in the order shown in the picture above. Then, show the order in which assignments are considered by **Backtracking** (this is analogous to the order in which nodes are expanded by depth-first search). Show the assignment even if it is immediately found to be inconsistent upon testing. Assume that the search continues even after finding a valid solution. Enter each assignment (one assignment in each of the spaces below) as a number from 1 to 4 (indicating the variable) followed by a letter, drawn from `R, G, B`, leave a space between them, e.g. `1 R`:
   a. _____
   b. _____
   c. _____
   d. _____
   e. _____

N.H.Reyes
Source: MIT OpenCourseware

f.
g.
h.
i.
j.

4. Repeat assuming we use **Backtracking with Forward Checking** to search for a solution to this problem. We use the same variable ordering and value ordering as before. Show the order in which assignments are considered by BT-FC. Whenever propagating after an assignment causes a domain to become empty, that causes backup in the search. Assume that the search continues even after finding a valid solution. Enter each assignment (one assignment in each of the spaces below) as before:

a.
b.
c.
d.
e.
f.
g.
h.
i.
j.

CSP Overview

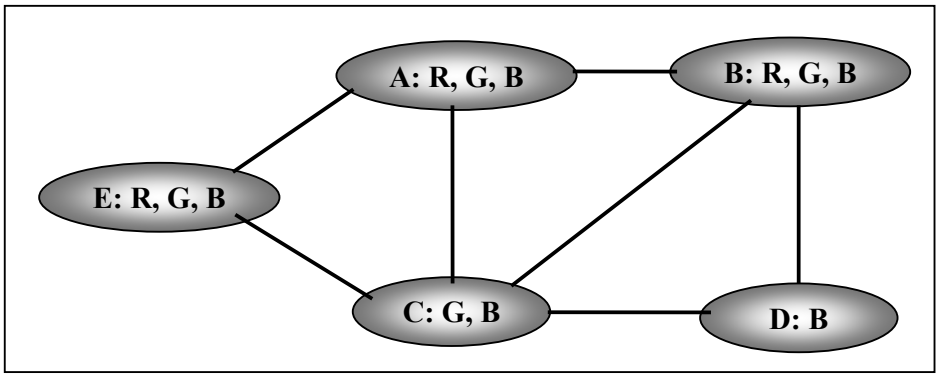
Indicate whether the following statements are true or false.

1. Constraint propagation without search is sufficient to find a satisfying set of assignments in most problems.
2. If constraint propagation leaves all variables with non-empty domains, there is at least one solution.
3. If constraint propagation leaves some variable with an empty domain, there is no solution.
4. If constraint propagation leaves all variables with singleton domains, there is a unique solution.
5. Constraint propagation takes on the order of ed^2 arc tests for one pass through all the arcs in a constraint graph with e edges and d values per domain.
6. Forward checking during backtracking requires verifying that the newly assigned value for a variable is consistent with all previously assigned values of variables.

7. Backtracking search with full constraint propagation (until arc-consistency is reached) generally results in an answer using fewer arc-tests than search with forward-checking only.
8. Backtracking with Forward Checking (BT-FC) never explores more possible variable values than pure backtracking.
9. BT-FC never tests more constraint arcs than pure backtracking.

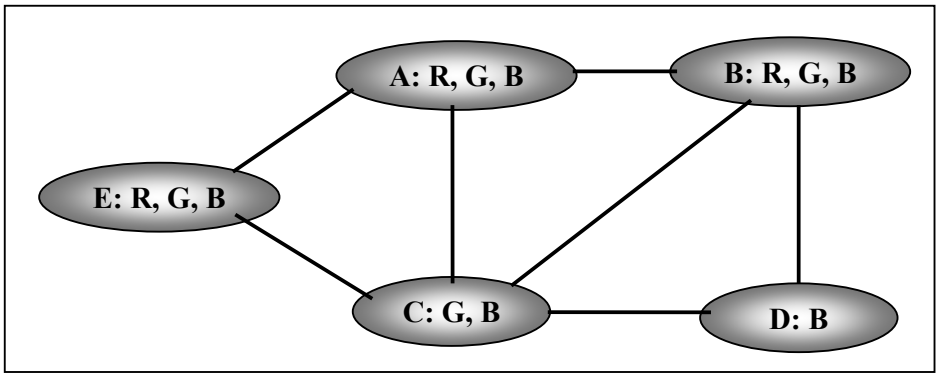
CSP Problems

Consider a map colouring problem consisting of five regions A, B, C, D and E. Your task is the standard map colouring problem. You are to assign each region a colour such that no two adjacent regions have the same colour. The legal colours are a subset of R, G and B, standing for Red, Green and Blue, respectively. The set of regions are depicted in the following constraint graph as ovals. The legal colours for each region is specified within its corresponding oval, and each pair of adjacent regions is denoted by a line between the two ovals corresponding to the two regions.



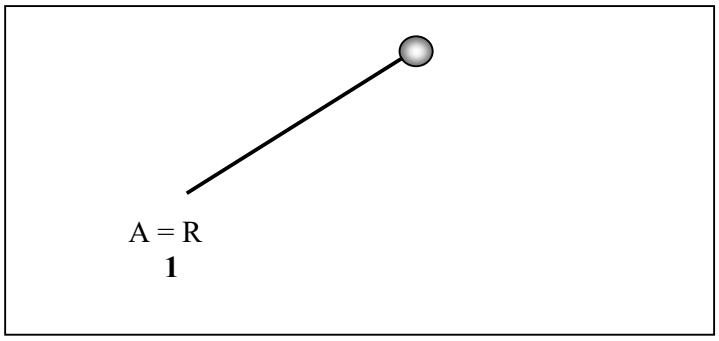
Part A. Pruned Domains

First, consider what domain elements can be eliminated using constraint propagation. Show your results on the copy of the constraint graph shown in the box below. On the graph, cross out each domain element that is eliminated by constraint propagation. For example, to start you off, we have eliminated **B**(lue) from region **B**, which is eliminated using constraint **B-D**.



Part B. Backtrack Search with Forward Checking

In this part you will use two CSP search methods to search for the first consistent solution to the map colouring problem described above. The two search methods you will use are **backtrack search** and **backtrack search with forward checking.** Use the same constraint graph.
Your task is to draw the search tree resulting from the application of each method, stopping when you reach the first solution, or the end of the search tree if no solution exists. In your tree you will label each node (except the root) with the variable/value assignment made at that point in the search (e.g. A = R). In addition, indicate the order in which assignments are made by writing a number under each assignment in your search tree. To get you going, we have started your first search tree below by indicating the root, and the first child expanded (labelled with assignment A=R and step number 1).



B.1. Simple Backtrack Search
Draw the search tree for simple backtrack search (no forward checking) in the box below. In your search, variables, denoting regions are to be ordered **alphabetically**, and colours are to be ordered **R**, then **G**, and finally **B**. Note that the full constraint propagation from Part A is **not** to be performed here.

B.2. Backtrack Search with Forward Checking
Draw the search tree for backtrack search with forward checking in the box below. In your search, variables, denoting regions are to be ordered **alphabetically**, and colours are to be ordered **R**, then **G**, and finally **B**. Note that the full constraint propagation from Part A is **not** to be performed here.

B.3. Backtrack Search with Dynamic Variable Ordering
Draw the search tree for backtrack search with dynamic variable ordering in the box below. Do not perform forward checking or full-constraint propagation. Select the variable to assign at each step based on the most constrained variable heuristic. Select the value of the variable to be assigned based on the least constrained value heuristic. We have not specified the assignment for the first step in this tree.